THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A row of triangles is made with sticks: 3 sticks for 1 triangle and 2 more sticks for each extra triangle. Sticks needed for 10 triangles: 1

(a) 21 **(b)** 30 **(c)** 23 **(d)** 20

2. A ray OB bisects $\angle AOC = 54^\circ$. Then $\angle AOB$ is: 1

(a) 54 **(b)** 27 **(c)** 63 **(d)** 108

3. The next palindrome after 1331 is: 1

(a) 1221 **(b)** 1441 **(c)** 1551 **(d)** 1342

4. In a pictograph one symbol = 5. Monday shows 4 symbols, Tuesday 6 symbols and Wednesday 3 symbols. The total number shown for the three days is: 1

(a) 60 **(b)** 13 **(c)** 70 **(d)** 65

5. The largest 3-digit multiple of 7 is: 1

(a) 987 **(b)** 980 **(c)** 994 **(d)** 999

6. A wire 40 cm long is bent into a rectangle 14 cm long. If it is bent into a square, its area (cm^2) is: 1

(a) 100 **(b)** 104 **(c)** 99 **(d)** 101

7. Which fraction lies between $\frac{1}{4}$ and $\frac{1}{2}$? 1

(a) $\frac{2}{4}$ **(b)** $\frac{2}{2}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{3}{8}$

8. A square has perimeter 32 cm. Its diagonal splits it into two triangles each with two sides of length (cm): 1

(a) 9 **(b)** 24 **(c)** 8 **(d)** 7

9. The sum of the number of lines of symmetry of a square and a regular hexagon is: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 12

10. A mountaineer is at 120 m above sea level and descends 150 m. His position (m) is: 1

(a) -30 **(b)** -28 **(c)** -20 **(d)** -32

11. The sum of the first 24 natural numbers added to the sum of the first 25 natural numbers is: 1

(a) 576 **(b)** 625 **(c)** 650 **(d)** 600

12. A ray is turned clockwise through a quarter turn and then through a further half turn (fractions of a full circle). The total angle turned is: 1

(a) 270 **(b)** 360 **(c)** 180 **(d)** 90

13. The sum of the first 30 counting numbers $1 + 2 + 3 + \cdots + 30$ is: 1

(a) 930 **(b)** 495 **(c)** 900 **(d)** 465

14. In the data 1, 4, 3, 5, 5, 1, 5, 3, 5, 5, 4, 5, 3, 3, 5, the value 6 occurs how many times? 1

(a) 3 **(b)** 2 **(c)** 0 **(d)** 1

15. How many prime numbers are there between 20 and 60? 1

(a) 11 **(b)** 8 **(c)** 10 **(d)** 9

16. A path of width 2 m is built inside a square garden of side 20 m along its boundary. The area of the remaining garden (m^2) is: 1

(a) 260 **(b)** 252 **(c)** 256 **(d)** 264

17. The numerator of $3\frac{2}{5}$ written as an improper fraction is: 1

(a) 17 **(b)** 22 **(c)** 18 **(d)** 16

18. Four equal circles of radius 3 cm are drawn touching in a row. The length of the row (cm) is: 1

(a) 24 **(b)** 36 **(c)** 27 **(d)** 21

19. How many of the letters in A, B, C, D, E have a line of symmetry? 1

(a) 7 **(b)** 4 **(c)** 6 **(d)** 5

20. Which pair gives the sum 5? 1

(a) -7 and 12 **(b)** 7 and 12 **(c)** -7 and 15 **(d)** -5 and 12

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Apply the reverse-and-add steps on 87 and state the palindrome reached. 2

22. In a pictograph, one symbol = 10 students. 5 symbols show Class VI-A. Class VI-B has 20 students more. How many symbols show VI-B? 2

23. Find the smallest number that must be added to 1234 to make it divisible by 9. 2

24. Find the cost of paving a rectangular courtyard 12 m by 8 m at ₹75 per m^2. 2

25. A number increased by $\frac{1}{2}$ of itself gives 18. Find the number. 2

26. In rectangle ABCD, $\angle AOB = 40^\circ$ at the point O where the diagonals meet. Find $\angle OAB$ and $\angle AOD$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Find the value of $2\frac{1}{2} + \frac{1}{3} - \frac{1}{4}$. 3

28. Three circles of radius 3 cm touch each other in a row. Find the distance between the centres of the first and third circles, and the total length of the row. 3

29. Write the order of rotational symmetry and number of lines of symmetry for the letters H, A, T. Which have more lines than the order? 3

30. A bank account has ₹300. Withdrawals of ₹200 and ₹250 are made, then ₹120 is deposited. Show the balance as an integer after each step and the final balance. 3

31. Write the 20th and 21st triangular numbers and check that they add to a square. 3

OR

There are 5 points on a line. How many line segments can be named? How many of them do not have A as an end point? 3

32. Find the sum of all numbers from 1 to 40 by pairing the first and last numbers. 3

OR

In a class of 15 students the data of the number of books read in a month is 3, 0, 1, 0, 4, 5, 4, 2, 1, 5, 2, 3, 3, 3, 1. How many read more than 2, and how many read exactly 2? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Compare a square and a regular octagon: find the angle of rotation of each, say which is larger, find the difference of the two angles, and the total number of lines of symmetry of the two figures together. 5

34. The temperatures (°C) at 5 places are -8, 3, -2, 11, -5. Arrange them from the coldest to the hottest, find the difference between the hottest and the coldest places and the sum of all temperatures. 5

35. A pattern of squares: the first figure has 3 squares and each next figure has 5 more. Find the number of squares in the 12th figure, and the first figure that has more than 108 squares. 5

36. A clock shows 2:00. The hour hand turns clockwise until the clock shows 7:00. Find the angle turned by the hour hand, the angle turned by the minute hand in this time, and the smaller angle between the hands at 7:00. 5

OR

A 4-digit palindrome has first digit 3 and digit sum 14. Find the number. Add 11 to it and see if the answer is a palindrome. 5

37. A pictograph (one symbol = 15) shows 4 symbols on Monday, 5 on Tuesday and 6 on Wednesday. On Thursday the number of items sold equals the total of Monday and Tuesday. Find the number sold on each day, and the total for the four days. 5

OR

A room is 540 cm long and 420 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

38. A rectangular hall 12 m by 9 m is to be covered with square tiles of side 40 cm costing ₹25 each. Find the number of tiles and the total cost. 5

OR

Ravi had ₹240. He spent $\frac{1}{6}$ of it on books and $\frac{2}{5}$ of the rest on a bag. How much does he have left? 5

END OF QUESTION PAPER — Total: 80 Marks